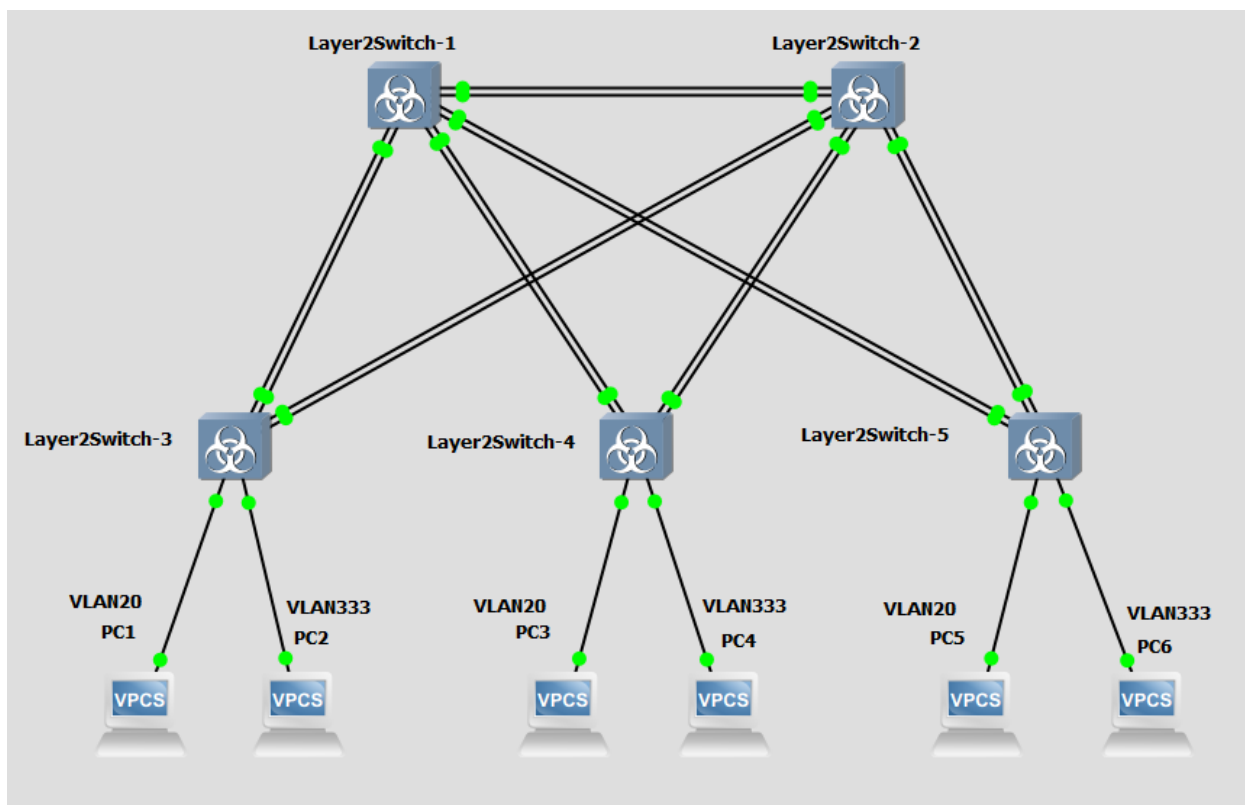


Тема: Настройка виртуальной локальной сети (VLAN)



- 1) Для заданной на схеме schema-lab3 сети, состоящей из управляемых коммутаторов и персональных компьютеров, настроить на коммутаторах логическую топологию используя протокол IEEE 802.1Q, для передачи пакетов VLAN333 между коммутаторами использовать Native VLAN.

На Layer2Switch-3:

Настраиваем порт на VPCS1:

```
Enable
conf t
vlan 20
vlan 333
exit
interface gi1/0
switchport mode access
switchport access vlan 20
no shutdown
```

Настраиваем порт на VPCS2:

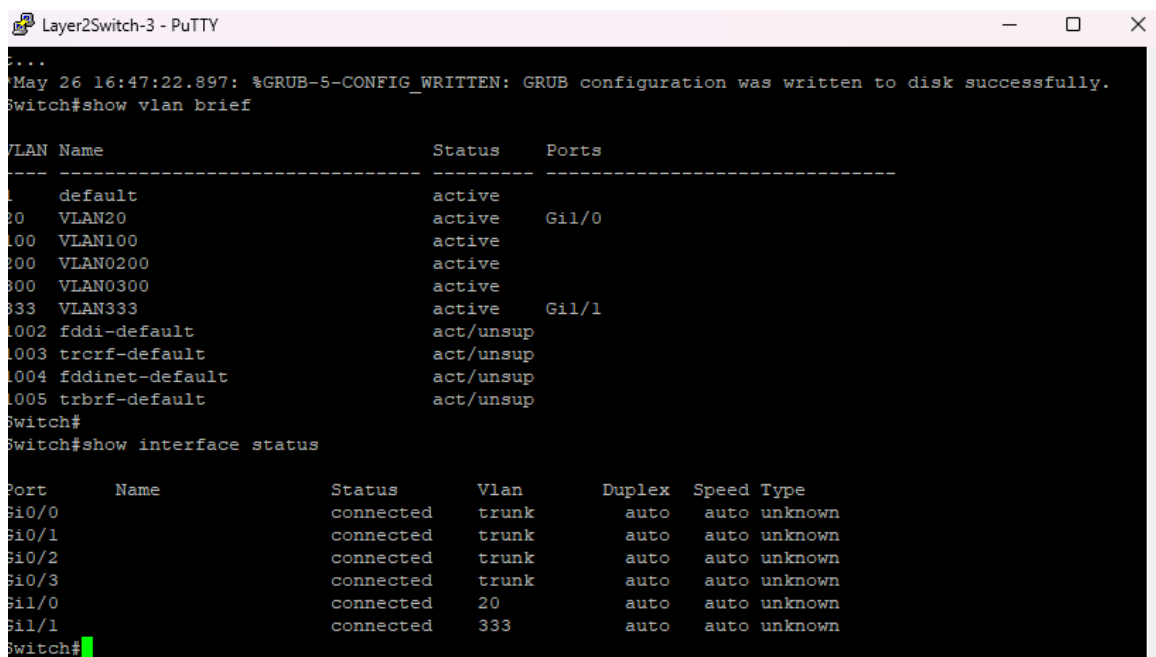
```
interface gi1/1  
  
switchport mode access  
  
switchport access vlan 322  
  
no shutdown
```

остальные порты:

```
interface range gi0/0-3  
  
switchport trunk encapsulation dot1q  
  
switchport mode trunk  
  
switchport trunk native vlan 333  
  
switchport trunk allowed vlan 20,333
```

write

Результат:



```
Layer2Switch-3 - PuTTY  
f...  
May 26 16:47:22.897: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.  
Switch#show vlan brief  
  
VLAN Name                Status    Ports  
-----  
1    default                 active      
20   VLAN20                  active    Gi1/0  
100  VLAN100                 active      
200  VLAN0200                active      
300  VLAN0300                active      
333  VLAN333                 active    Gi1/1  
1002 fddi-default            act/unsup  
1003 trcrf-default         act/unsup  
1004 fddinet-default       act/unsup  
1005 trbrf-default         act/unsup  
Switch#  
Switch#show interface status  
  
Port      Name                Status    Vlan    Duplex  Speed  Type  
-----  
Gi0/0     connected          trunk     auto    auto    unknown  
Gi0/1     connected          trunk     auto    auto    unknown  
Gi0/2     connected          trunk     auto    auto    unknown  
Gi0/3     connected          trunk     auto    auto    unknown  
Gi1/0     connected          20       auto    auto    unknown  
Gi1/1     connected          333      auto    auto    unknown  
Switch#
```

Настройка на 4 и 5 свитче аналогична.

Layer2Switch-4:

```
Layer2Switch-4 - PuTTY
Switch#write
Building configuration...
Compressed configuration from 5589 bytes to 2159 bytes[OK]
*May 26 16:51:11.371: %GRUB-S-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
Switch#
*May 26 16:51:12.827: %GRUB-S-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Switch#
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active
20   VLAN20                  active    Gi1/0
100  VLAN100                  active
200  VLAN0200                 active
300  VLAN0300                 active
333  VLAN333                  active    Gi1/1
1002 fddi-default             act/unsup
1003 trcrf-default          act/unsup
1004 fddinet-default         act/unsup
1005 trbrf-default          act/unsup
Switch#show interface status

Port      Name      Status    Vlan    Duplex  Speed Type
-----
Gi0/0     connected trunk      auto    auto unknown
Gi0/1     connected trunk      auto    auto unknown
Gi0/2     connected trunk      auto    auto unknown
Gi0/3     connected trunk      auto    auto unknown
Gi1/0     connected 20        auto    auto unknown
Gi1/1     connected 333       auto    auto unknown
Switch#
```

Layer2Switch-5:

```
Layer2Switch-5 - PuTTY
Switch#
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active
20   VLAN20                  active    Gi1/0
100  VLAN100                  active
200  VLAN0200                 active
300  VLAN0300                 active
333  VLAN333                  active    Gi1/1
1002 fddi-default             act/unsup
1003 trcrf-default          act/unsup
1004 fddinet-default         act/unsup
1005 trbrf-default          act/unsup
Switch#show interface status

Port      Name      Status    Vlan    Duplex  Speed Type
-----
Gi0/0     connected trunk      auto    auto unknown
Gi0/1     connected trunk      auto    auto unknown
Gi0/2     connected trunk      auto    auto unknown
Gi0/3     connected trunk      auto    auto unknown
Gi1/0     connected 20        auto    auto unknown
Gi1/1     connected 333       auto    auto unknown
Switch#
```

На Layer2Switch-1 и Layer2Switch-2 все порты – trunk порты:

На всех портах:

switchport trunk native vlan 333

switchport trunk allowed vlan 20,333

```
Layer2Switch-1 - PuTTY

Switch#show interface trunk

Port      Mode      Encapsulation  Status      Native vlan
Gi0/0     on        802.1q         trunking    333
Gi0/1     on        802.1q         trunking    333
Gi0/2     on        802.1q         trunking    333
Gi0/3     on        802.1q         trunking    333
Gi1/0     on        802.1q         trunking    333
Gi1/1     on        802.1q         trunking    333
Gi1/2     on        802.1q         trunking    333
Gi1/3     on        802.1q         trunking    333

Port      Vlans allowed on trunk
Gi0/0     20,333
Gi0/1     20,333
Gi0/2     20,333
Gi0/3     20,333
Gi1/0     20,333
Gi1/1     20,333
Gi1/2     20,333
Gi1/3     20,333

Port      Vlans allowed and active in management domain
Gi0/0     20,333
--More--
```

```
Layer2Switch-2 - PuTTY

1004 fddinet-default      act/unsup
1005 trbrf-default        act/unsup
Switch#
Switch#
Switch#show interfaces trunk

Port      Mode      Encapsulation  Status      Native vlan
Gi0/0     on        802.1q         trunking    333
Gi0/1     on        802.1q         trunking    333
Gi0/2     on        802.1q         trunking    333
Gi0/3     on        802.1q         trunking    333
Gi1/0     on        802.1q         trunking    333
Gi1/1     on        802.1q         trunking    333
Gi1/2     on        802.1q         trunking    333
Gi1/3     on        802.1q         trunking    333

Port      Vlans allowed on trunk
Gi0/0     20,333
Gi0/1     20,333
Gi0/2     20,333
Gi0/3     20,333
Gi1/0     20,333
Gi1/1     20,333
Gi1/2     20,333
Gi1/3     20,333

Port      Vlans allowed and active in management domain
Gi0/0     20,333
Gi0/1     20,333
--More--
```

- 2) Проверить доступность персональных компьютеров, находящихся в одинаковых VLAN и недоступность находящихся в различных, результаты задокументировать.

Ip адреса:

Маска 255.255.255.0

VLAN 20:

PC1 - 192.168.1.2

PC3 - 192.168.1.3

PC5 - 192.168.1.4

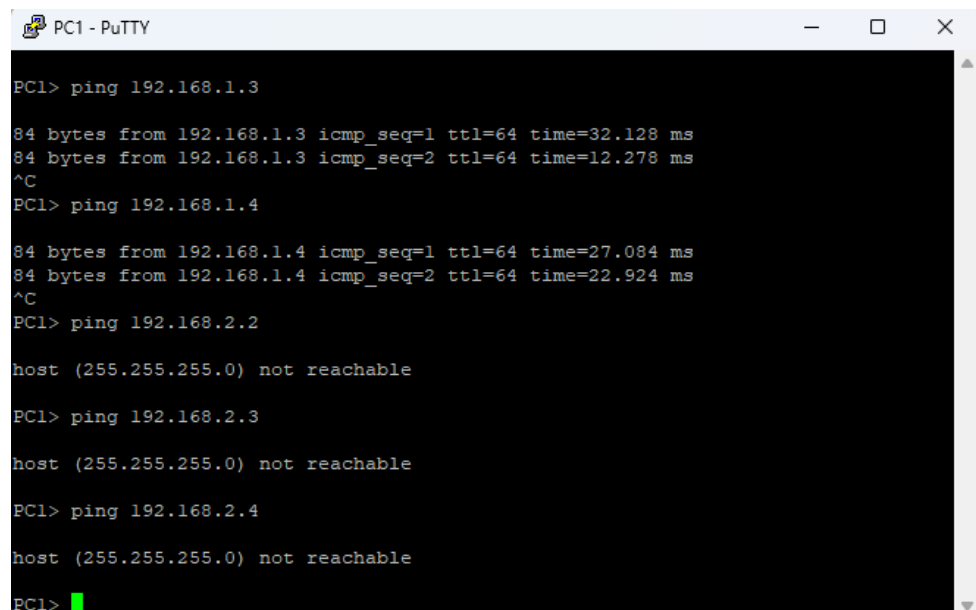
VLAN 333:

PC2 - 192.168.2.2

PC4 - 192.168.2.3

PC6 - 192.168.2.4

Пинги:



```
PC1> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=32.128 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=12.278 ms
^C
PC1> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=27.084 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=22.924 ms
^C
PC1> ping 192.168.2.2

host (255.255.255.0) not reachable

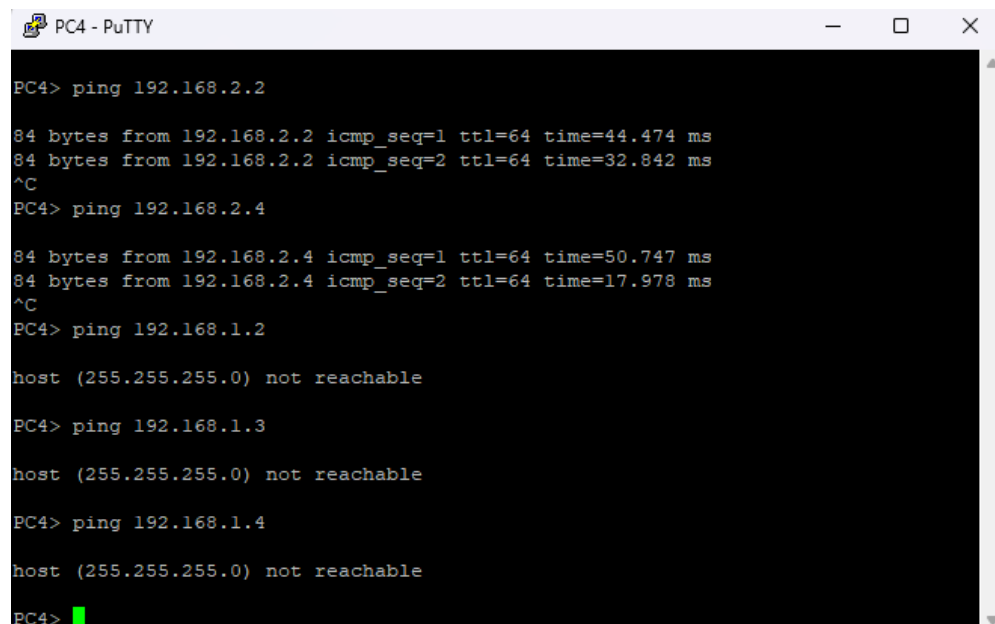
PC1> ping 192.168.2.3

host (255.255.255.0) not reachable

PC1> ping 192.168.2.4

host (255.255.255.0) not reachable

PC1>
```



```
PC4> ping 192.168.2.2

84 bytes from 192.168.2.2 icmp_seq=1 ttl=64 time=44.474 ms
84 bytes from 192.168.2.2 icmp_seq=2 ttl=64 time=32.842 ms
^C
PC4> ping 192.168.2.4

84 bytes from 192.168.2.4 icmp_seq=1 ttl=64 time=50.747 ms
84 bytes from 192.168.2.4 icmp_seq=2 ttl=64 time=17.978 ms
^C
PC4> ping 192.168.1.2

host (255.255.255.0) not reachable

PC4> ping 192.168.1.3

host (255.255.255.0) not reachable

PC4> ping 192.168.1.4

host (255.255.255.0) not reachable

PC4>
```

- 3) Перехватить в Wireshark пакеты с тегами и без тегов (nb!), результаты задокументировать.

Пакет от PC1 до Layer2Switch-3

No.	Time	Source	Destination	Protocol	Length	Info
21	41.855157	192.168.1.2	192.168.1.3	ICMP	98	Echo (ping) request id=0x1cd4, seq=1/256, ttl=64 (reply in 22)
22	41.905551	192.168.1.3	192.168.1.2	ICMP	98	Echo (ping) reply id=0x1cd4, seq=1/256, ttl=64 (request in 21)
24	42.906236	192.168.1.2	192.168.1.3	ICMP	98	Echo (ping) request id=0x1dd4, seq=2/512, ttl=64 (reply in 25)
25	42.947252	192.168.1.3	192.168.1.2	ICMP	98	Echo (ping) reply id=0x1dd4, seq=2/512, ttl=64 (request in 24)
26	43.948304	192.168.1.2	192.168.1.3	ICMP	98	Echo (ping) request id=0x1ed4, seq=3/768, ttl=64 (reply in 27)
27	43.982303	192.168.1.3	192.168.1.2	ICMP	98	Echo (ping) reply id=0x1ed4, seq=3/768, ttl=64 (request in 26)
31	44.982521	192.168.1.2	192.168.1.3	ICMP	98	Echo (ping) request id=0x1fd4, seq=4/1024, ttl=64 (reply in 32)
32	45.013305	192.168.1.3	192.168.1.2	ICMP	98	Echo (ping) reply id=0x1fd4, seq=4/1024, ttl=64 (request in 31)

```
Frame 21: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
  Section number: 1
    Interface id: 0 (-)
    Encapsulation type: Ethernet (1)
    Arrival Time: May 27, 2026 00:10:52.866891000 Новосибирское стандартное время
    UTC Arrival Time: May 26, 2026 17:10:52.866891000 UTC
    Epoch Arrival Time: 1779815452.866891000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 727.000 microseconds]
    [Time since reference or first frame: 41.855157000 seconds]
    Frame Number: 21
    Frame Length: 98 bytes (784 bits)
    Capture Length: 98 bytes (784 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:icmp:data]
    Character encoding: ASCII (0)
    [Coloring Rule Name: ICMP]
    [Coloring Rule String: icmp || icmpv6]
  Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Private_66:68:02 (00:50:79:66:68:02)
    Destination: Private_66:68:02 (00:50:79:66:68:02)
      ....0. .... = LG bit: Globally unique address (factory default)
      ....0. .... = IG bit: Individual address (unicast)
    Source: Private_66:68:00 (00:50:79:66:68:00)
      Type: IPv4 (0x0800)
      [Stream index: 4]
    Internet Protocol Version 4, Src: 192.168.1.2, Dst: 192.168.1.3
    Internet Control Message Protocol
```

Не тегирован.

От Layer2Switch-3 до Layer2Switch-1:

```
Захват с Standard input (Layer2Switch-1 Ethernet2 to Layer2Switch-3 Ethernet0)
icmp
No. Time Source Destination Protocol Length Info
21 24.082729 192.168.1.2 192.168.1.3 ICMP 102 Echo (ping) request id=0x1cd4, seq=1/256, ttl=64 (reply in 22)
22 24.123748 192.168.1.3 192.168.1.2 ICMP 102 Echo (ping) reply id=0x1cd4, seq=1/256, ttl=64 (request in 21)
25 25.126090 192.168.1.2 192.168.1.3 ICMP 102 Echo (ping) request id=0x1dd4, seq=2/512, ttl=64 (reply in 26)
26 25.161099 192.168.1.3 192.168.1.2 ICMP 102 Echo (ping) reply id=0x1dd4, seq=2/512, ttl=64 (request in 25)
28 26.176744 192.168.1.2 192.168.1.3 ICMP 102 Echo (ping) request id=0x1ed4, seq=3/768, ttl=64 (reply in 29)
29 26.199026 192.168.1.3 192.168.1.2 ICMP 102 Echo (ping) reply id=0x1ed4, seq=3/768, ttl=64 (request in 28)
32 27.205161 192.168.1.2 192.168.1.3 ICMP 102 Echo (ping) request id=0x1fd4, seq=4/1024, ttl=64 (reply in 33)
33 27.232000 192.168.1.3 192.168.1.2 ICMP 102 Echo (ping) reply id=0x1fd4, seq=4/1024, ttl=64 (request in 32)

Frame 21: Packet, 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface -, id 0
Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Private_66:68:02 (00:50:79:66:68:02)
  Destination: Private_66:68:02 (00:50:79:66:68:02)
    ....0. .... = LG bit: Globally unique address (factory default)
    ....0. .... = IG bit: Individual address (unicast)
  Source: Private_66:68:00 (00:50:79:66:68:00)
    Type: 802.1Q Virtual LAN (0x8100)
    [Stream index: 5]
  802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
    000. .... = Priority: Best Effort (default) (0)
    ...0. .... = DEI: Ineligible
    ...0000 0001 0100 = ID: 20
    Type: IPv4 (0x0800)
  Internet Protocol Version 4, Src: 192.168.1.2, Dst: 192.168.1.3
  Internet Control Message Protocol
    Type: Echo (ping) request (8)
```

Тегирован, vlan_id =20.

Пинг с PC2 до PC4:

